Ashfall

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For the term project I have decided to do a simulation of wildfires. Living in the Northwest and dealing with wildfires is one of the inspirations behind this idea. I've also been fascinated by how wildfires create fertile ground for new life to grow, so I'm hoping to capture a little bit of that life and death cycle in this simulation.

In order to simulate a wildfire my plan is to have a grid of tiles where each tile represents one of 4 states. A green tile will represent a living tree, a brown tile will represent a dead tree, an orange tile will represent a burning tree, and a black tile will represent the ashes of a burned tree. I have a little experience drawing basic shapes with graphics libraries, so the plan is to display the simulation in a separate window and not in a shell. The program will follow some simple rules to determine transitions between different states. A living tree will have a chance to turn into a dead tree, increasing with the number of living trees in surrounding 8 tiles. A dead tree will have a chance to catch on fire, increasing with the number of surrounding dead trees. Burning trees will always catch adjacent dead trees on fire and have a chance to catch live trees. Burning trees will turn to ash. Ash tiles will have a chance to turn into live tiles, increasing with the number of live tiles surrounding it.

While these rules aren't as simple as something like Conway's Game of Life, I think they're simple enough to fit the project's parameters. I'm also hoping that these rules will create some pretty cool behaviors. I think there should be some boom-and-bust cycles where large forests are created, followed by larger and larger dead patches, which should result in large fires occurring. I'm also hoping that there's a good variety in the size of fires, with mostly small fires and the occasional large one. The last thing that I'm looking for is for a dynamic mosaic effect, where there are patches of green, brown, orange, and black that are all changing with each other.

The biggest concern I have about the project is trying to find the right probabilities for state changes. I think there will be a sweet spot for the probability of each state change in which interesting behaviors emerge. It could require a ton of testing to figure out though, so it could add a good chunk of time to the project. I also have a minor concern about the graphics since I don't have a ton of experience, but I have strong math skills so I think I'll be able to figure it out.